

A Non-Invasive Acoustic Sensing System for Bruxism Screening

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Overview

What is Bruxism?

Research Objective

Methods

Results

What does this mean for Us?

Conclusions

What is Bruxism?

Oral Disorder: Grinding/Clenching of the Teeth

Globally (sleep & awake):

24%

ADULTS

19%

CHILDREN

Globally (sleep):

21%

Overall

Causes:

- Stress/anxiety
- Tooth alignment
- Medication

Symptoms:

- Tooth wear
- Pain/stiffness in jaw
- Headaches
- Poor Sleep

Research Objective

Can we accurately detect bruxism using acoustic signals from a patient overnight?

Can this be accomplished through a low-cost wearable?

Benefits:

- Early detection
- Enable remote monitoring
- Make screening accessible
- Help track medication effects

What are current solutions?

Polysomnography (PSG) or Commercial Devices:

	Bitestrip	dia-BRUXO	GrindCare	Bruxoff	NOX T3	Sleep Profiler	Earable
EMG-based	X	X	X	X	X	X	
Audio recording					X	X	X
Multi-night capable			X	X			X
Sensitivity / Specificity vs PSG-AV	72–84.2% / —	not evaluated	60% / 60%	83.3% / 72.2%	48.3% / 81.2%	100% / 93.3%	not evaluated

Source: Australian Dental Journal, 2024

Supporting Research?

Signal Capture in the Ear:

"The occlusion effect is commonly experienced by in-ear device wearers as an increased loudness sensation of bone-conducted low frequency sounds." (Acta Acustica, 2021)

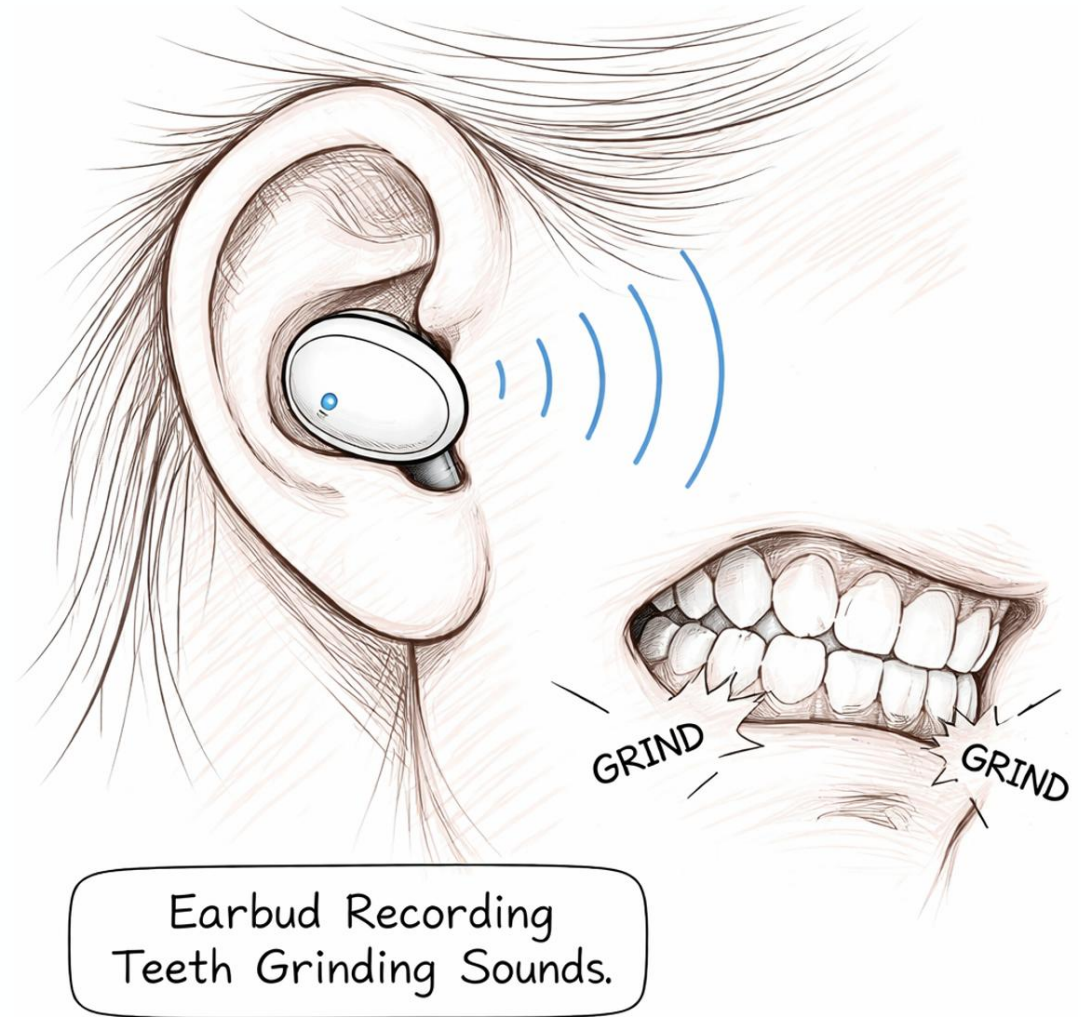
Acoustic Approach for Bruxism Detection:

A device is "configured to sense acoustical signals generated by teeth grinding" and "perform signal processing to differentiate between teeth grinding and other acoustical signals" (US Patent, 2025)

Filtering for Teeth Grinding:

"A Study on the Detection of Sleep Bruxism Using Acoustic Features " found that snoring produces acoustic signatures distinguishable from bruxism. (PubMed, 2025)

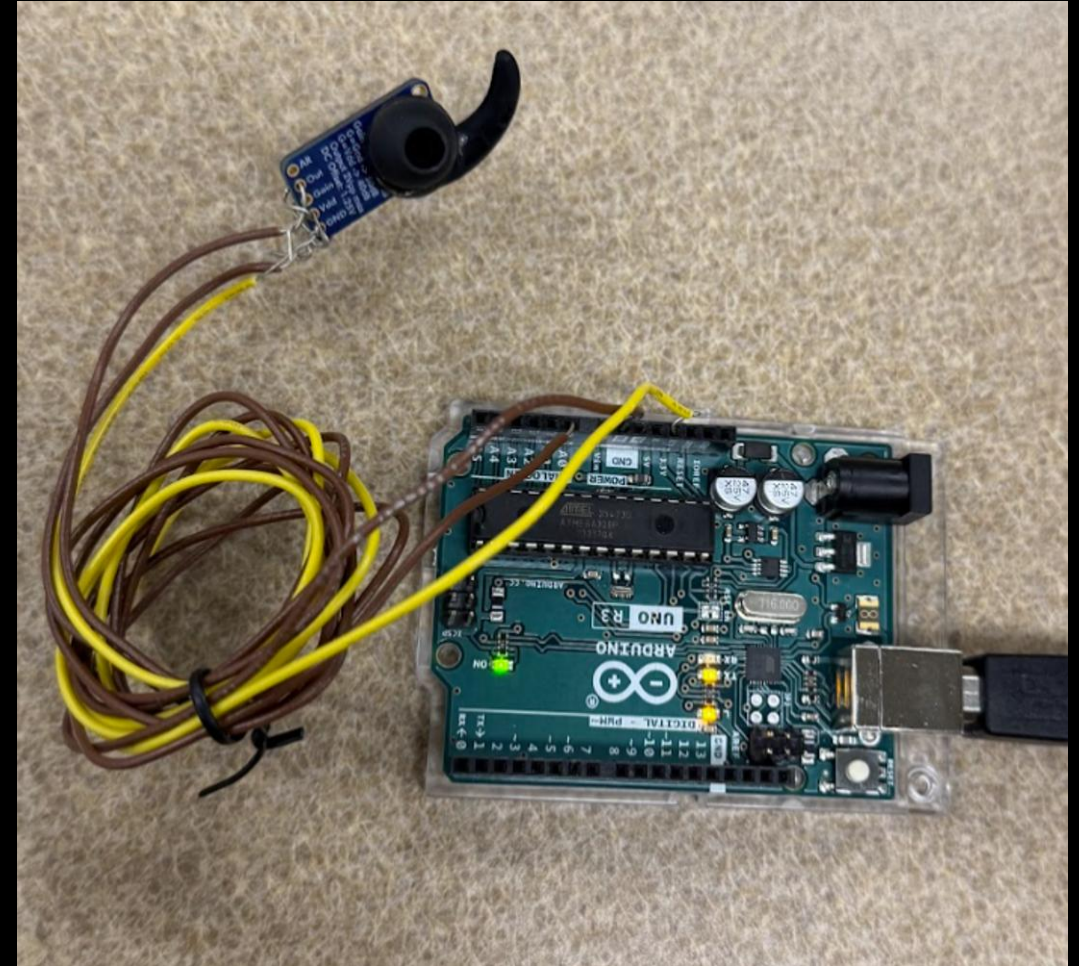
Prototype VS Final Project



Methods

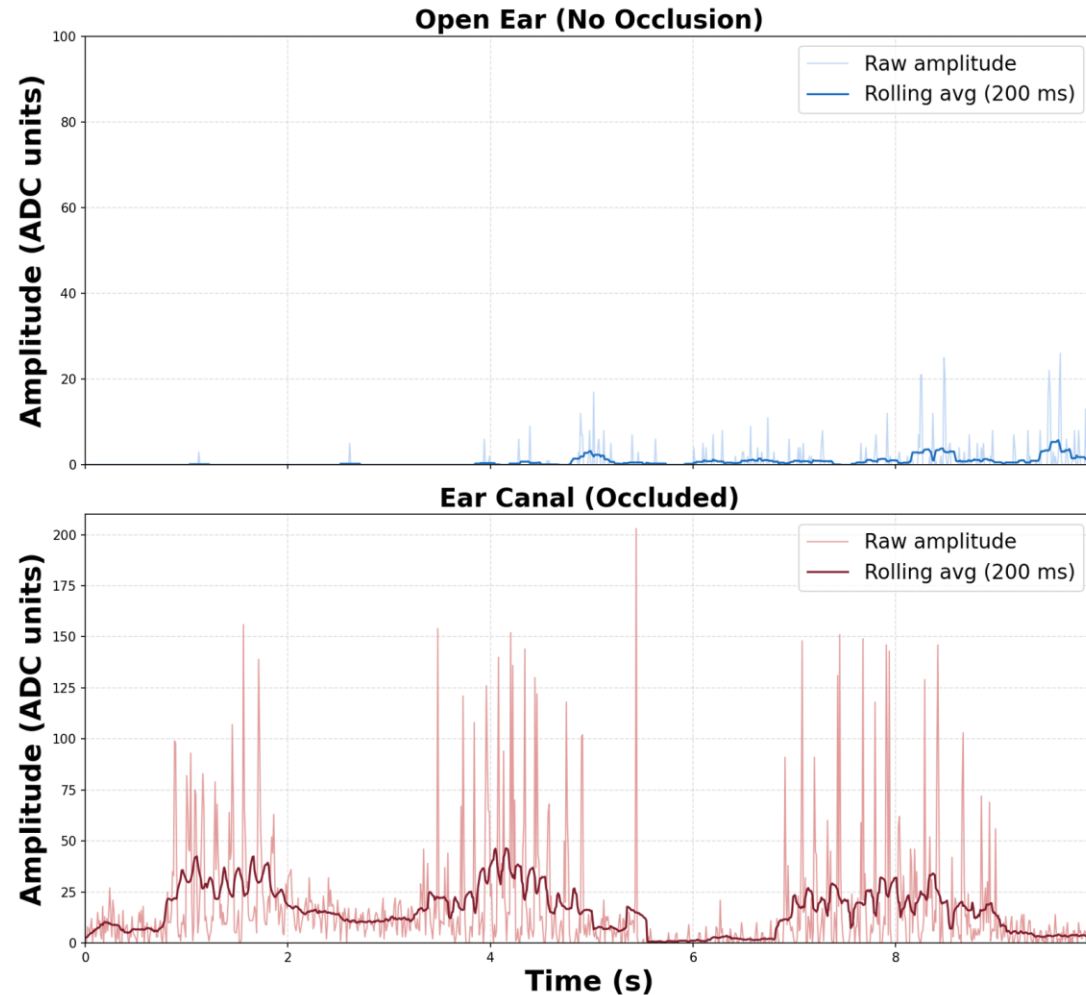
Sample Size: N=1 (The researcher)

- Hardware: Arduino and MEMS microphone
- Physiologically mimicked teeth grinding
- Controlled Settings
- Excel: Store Data
- MATLAB for Data Analysis and Graphing

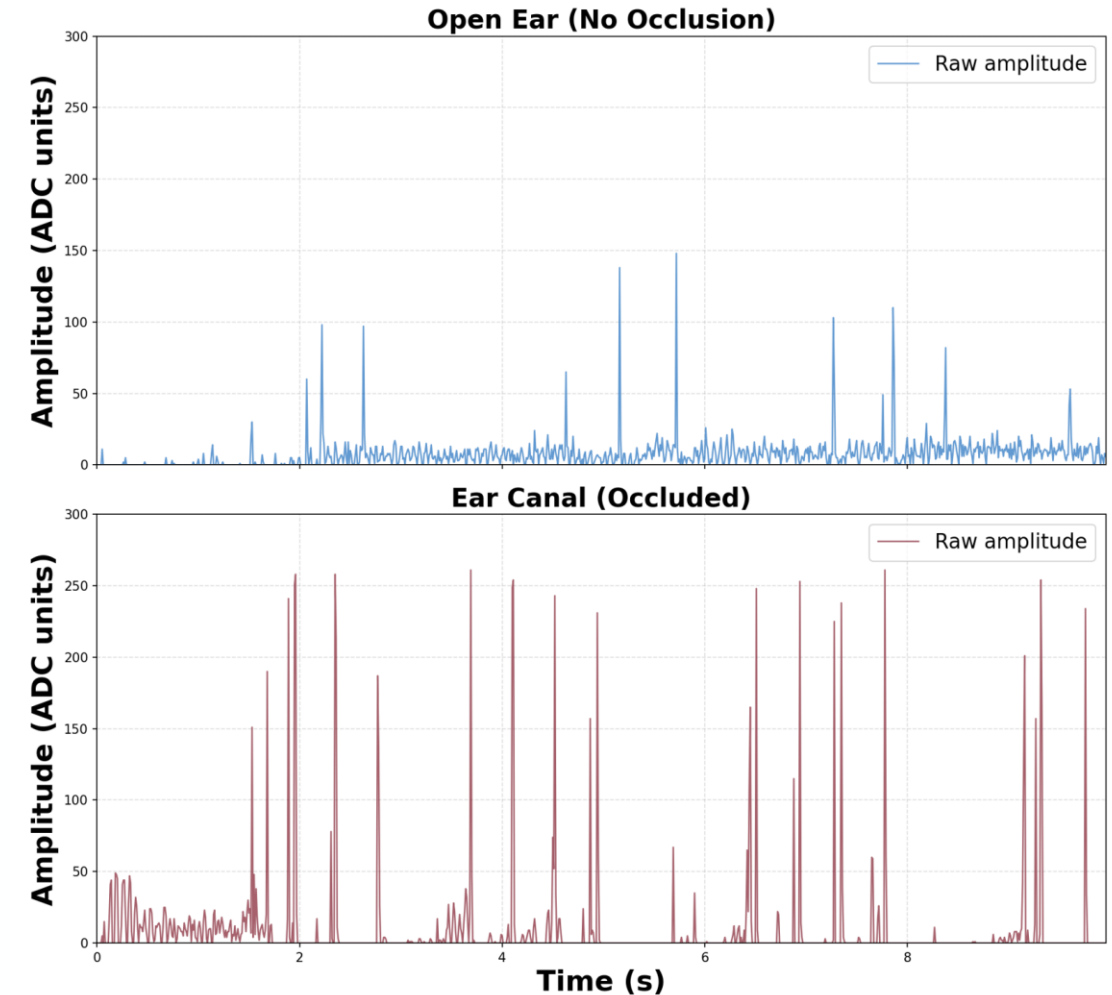


Occlusion Effect Trial:

Teeth Grinding: Open Ear vs. Occluded (Ear Canal)

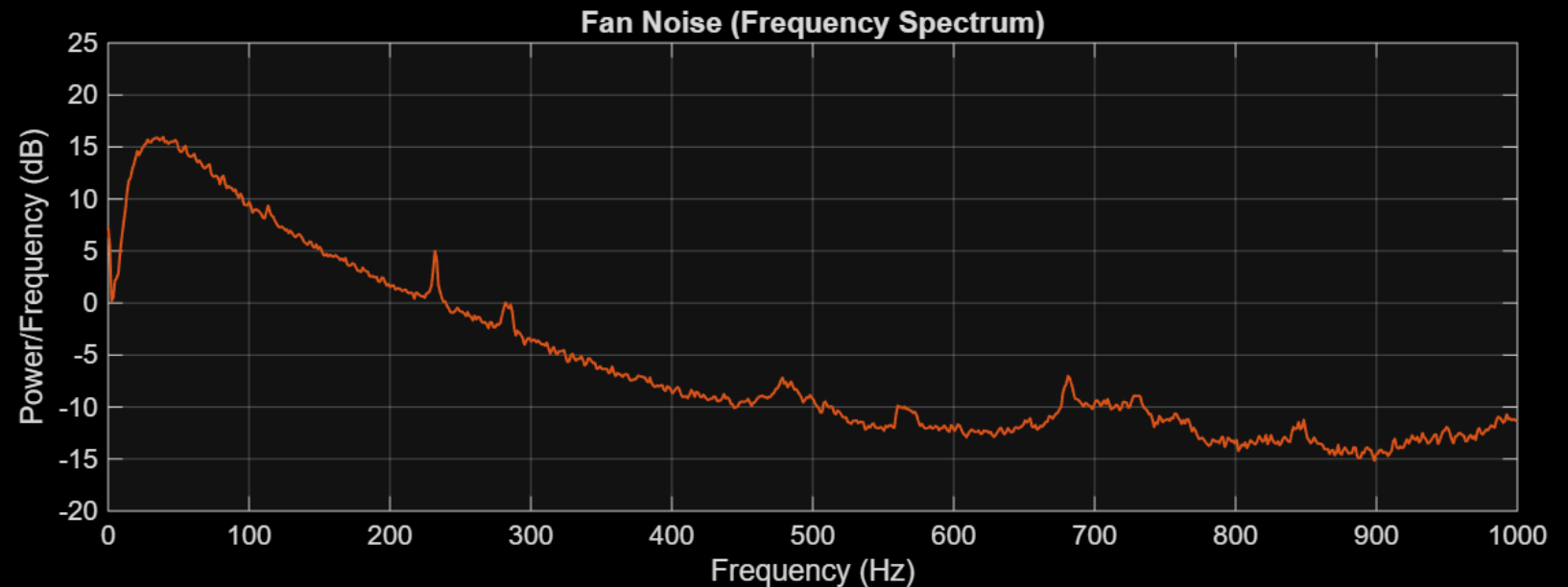
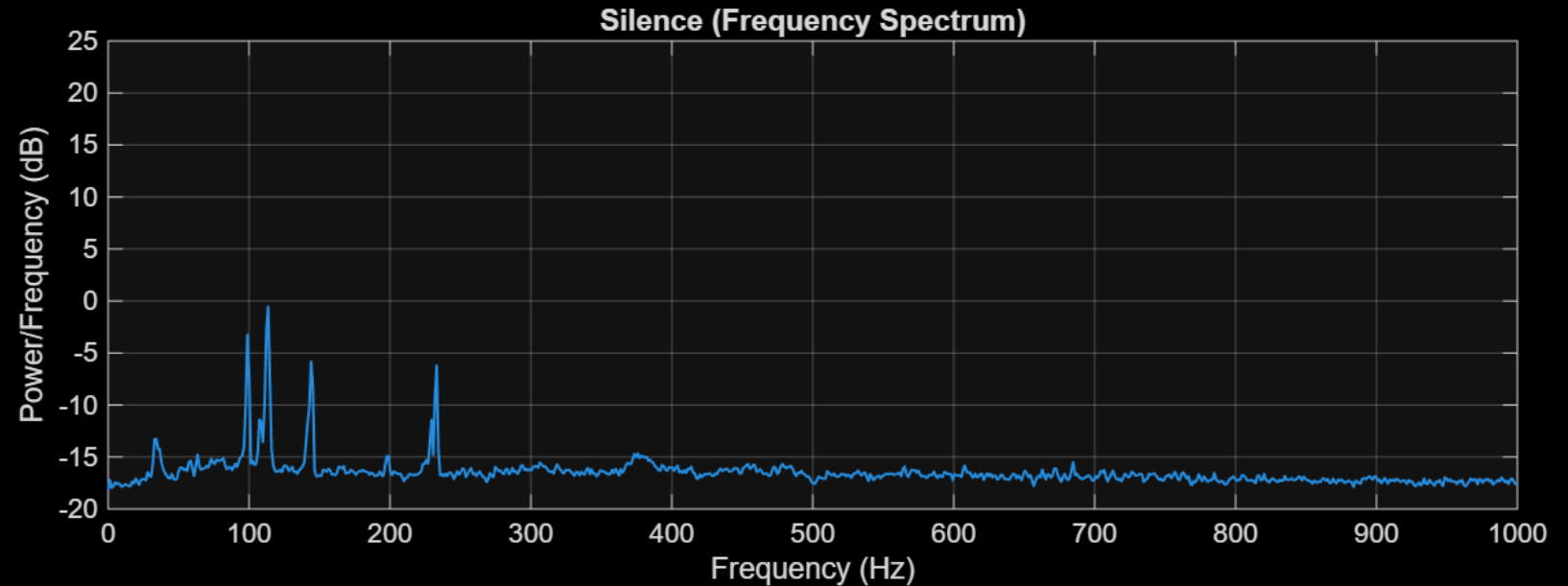


Teeth Snapping: Open Ear vs. Occluded (Ear Canal)

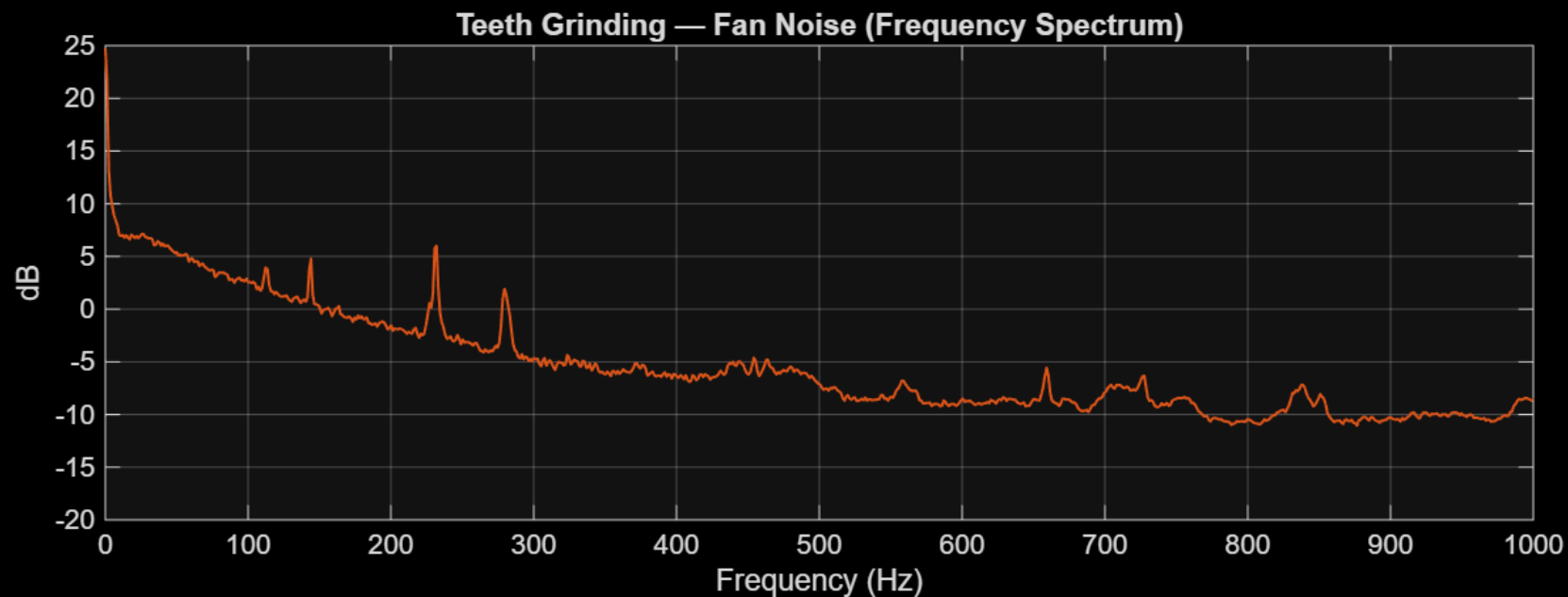
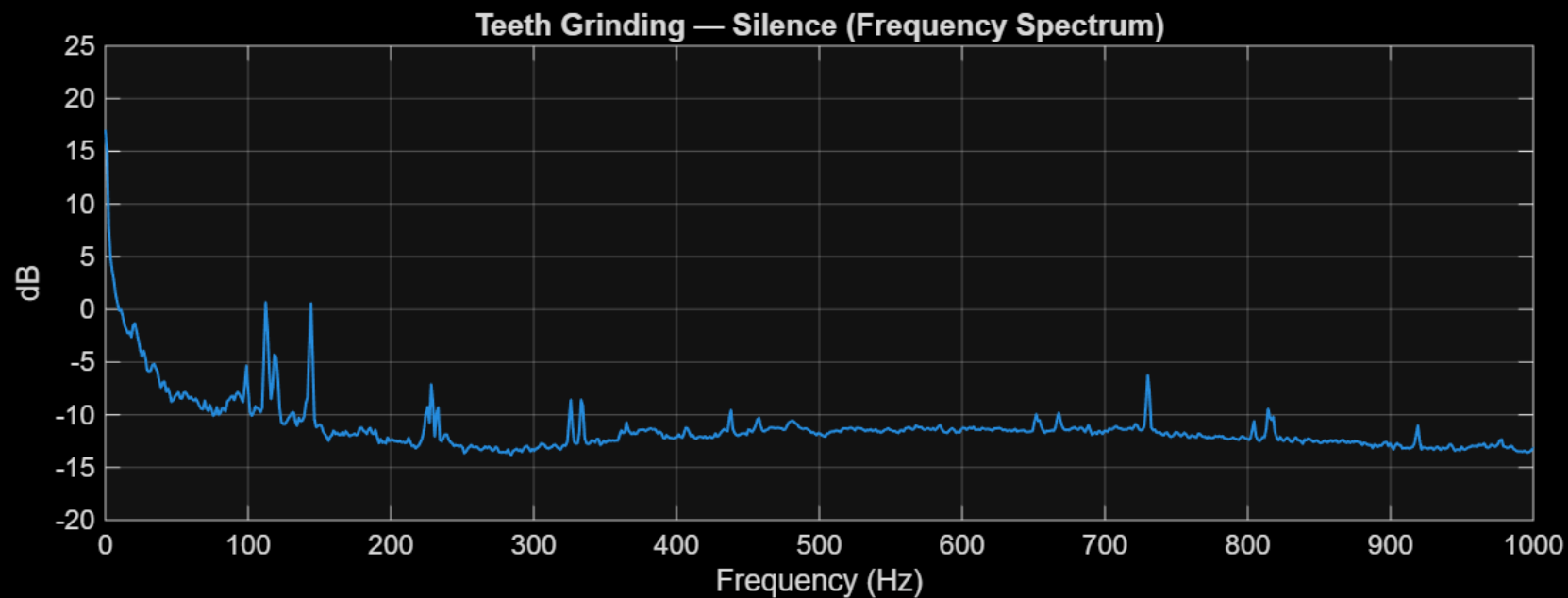


Acoustic Approach: Raw Silence and Fan

Bruxism Signal: Silence vs Fan Noise — Dominant Frequency Comparison



Bruxism Signal: Silence vs Fan Noise — Dominant Frequency Comparison



**Acoustic
Approach:
Teeth Grinding**

Discussion

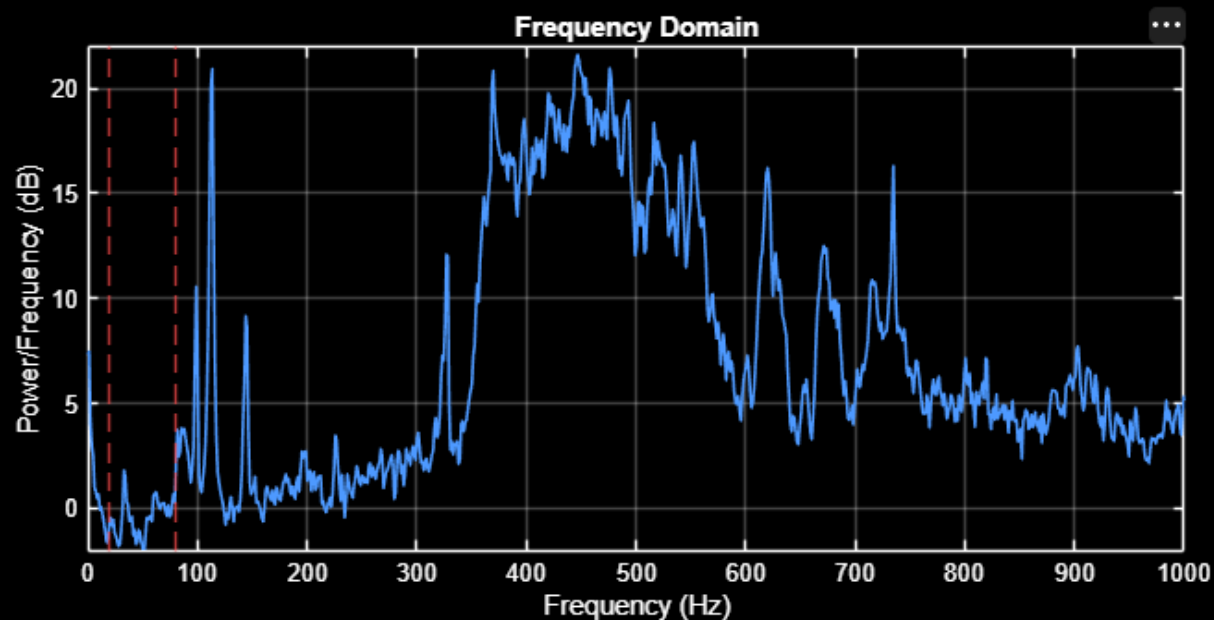
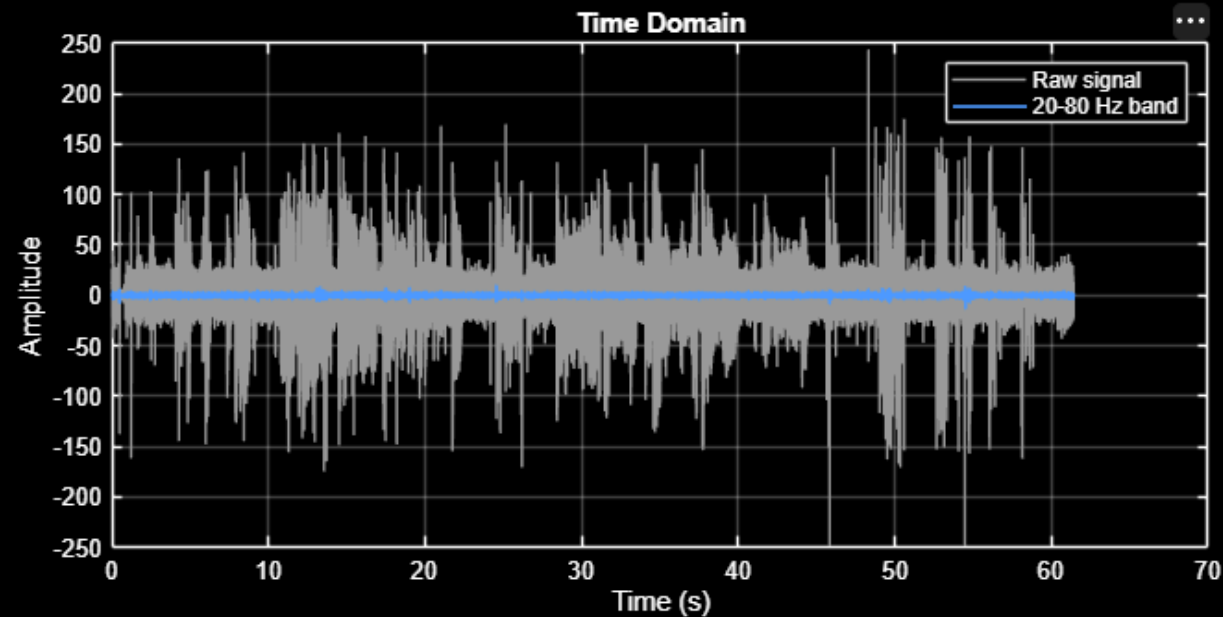
OCCLUSION EFFECT:

- Occlusion Effect increased signal capture by ~10x
- Useful Strategy

ACOUSTIC APPROACH:

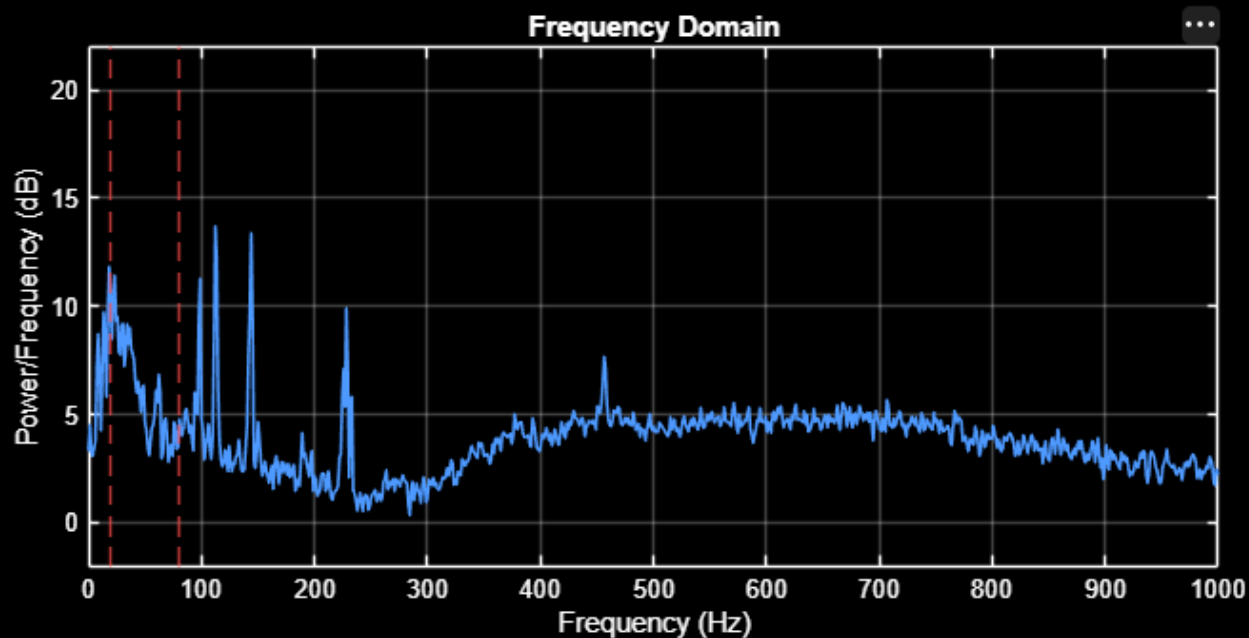
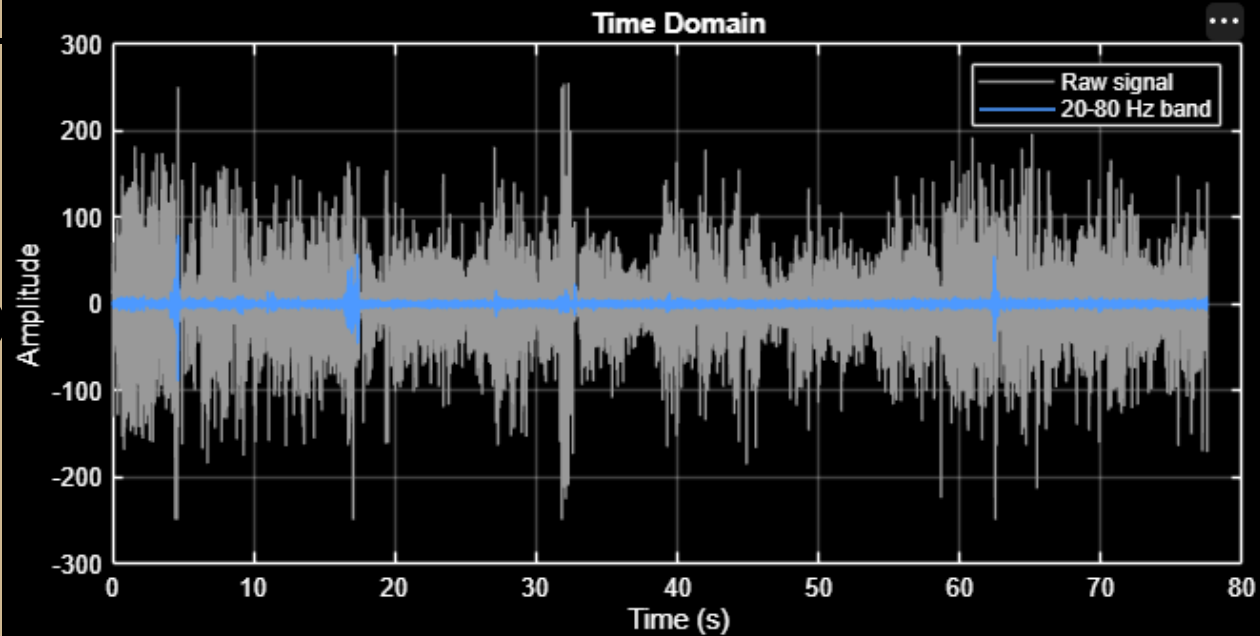
- Silence baseline: ~110, 140, 230 Hz
- Fan baseline: ~110, 140, 230, 285 Hz
- Grinding-in-silence: ~110, 140, 230, 285 Hz
- Grinding-in-fan: ~110, 140, 230 Hz
- **Possible range for teeth grinding: [20-80 Hz]**
- **Fan noise interfered with grinding detection**
stronger occlusion needed

Filtering Algorithm



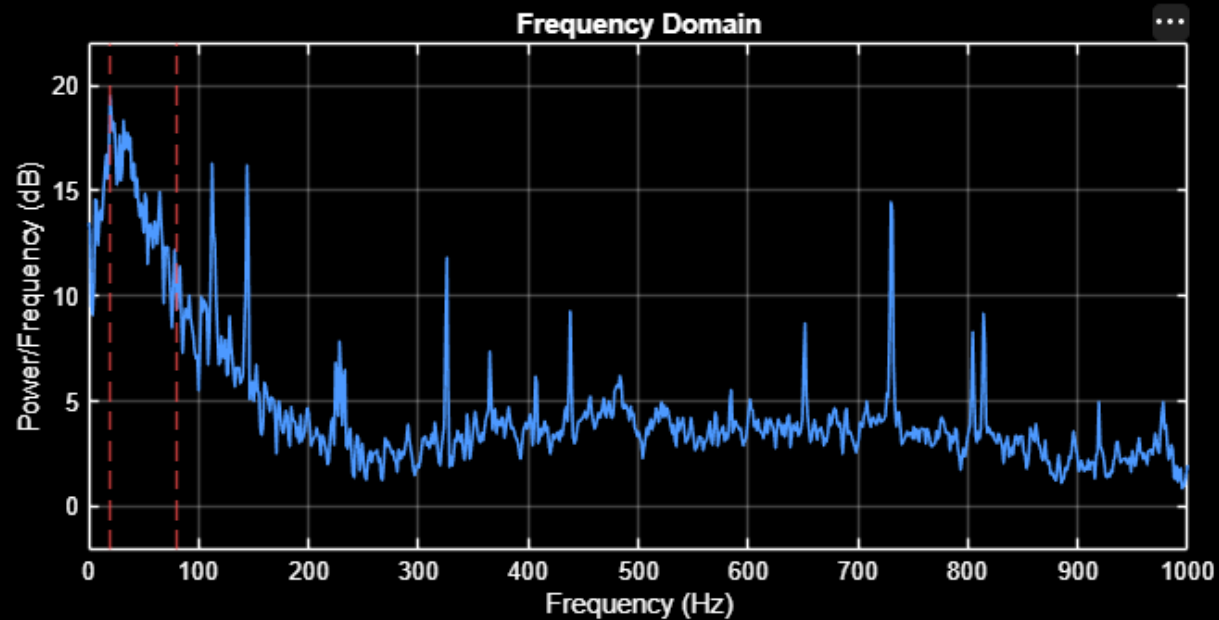
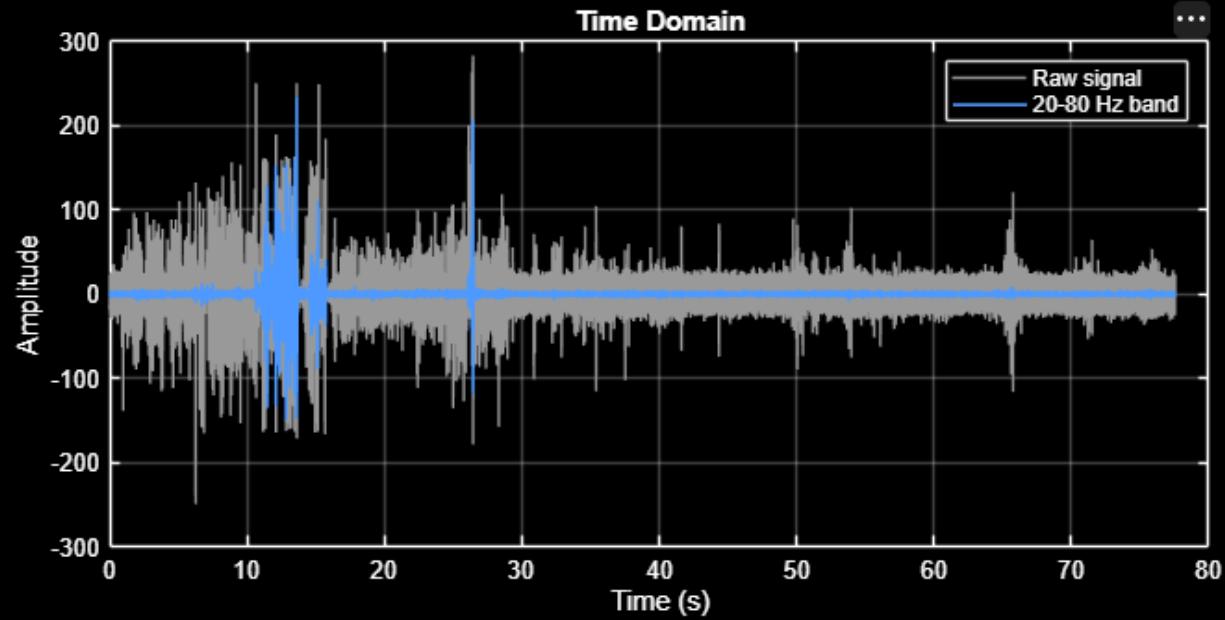
No
Warning

Filtering Algorithm



Mild
Warning

Filtering Algorithm



Severe
Warning

Discussion

	No Warning	Mild	Severe
Accuracy	3/3	3/3	3/3
Total Active Time in Grinding Band	32.5 s	12.0 s	35.8 s
Number of Bursts Clearing Mild Length Floor (≥ 0.25 s)	0.3	7.0	10.7
Number of Bursts Clearing Severe Length Floor (≥ 1.5 s)	0	0.3	2.7
Longest Continuous Grinding Activity	0.16 s	1.39 s	4.04s

Shortcomings/Challenges

- Sample size
- Ear-bud Fit
- No detect non-acoustic clenching
- Filtering out movement artifacts
- No Sensitivity and Specificity Data

What's Next?

- Consider new findings
- Use pattern recognition algorithms / machine learning
- Run clinical tests in natural conditions to assess feasibility beyond controlled lab testing

THANK YOU

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Terms

Fourier Transform: converts signal into its frequency components, showing frequencies present and their amplitudes

MEMS: Micro-Electro-Mechanical Systems

Sensitivity: how well your system detects true events

Specificity: how well your system avoids false alarms

Harmonic: a frequency that is an integer multiple of a fundamental frequency (ex. 60 (fundamental), 120, 180 Hz)

Broadband: signal containing many different frequencies at the same time without a specific pitch

Tonal: a sound built around a single clear frequency